A Term Project Report

On

Sustainable UV-C Hygiene Chamber to Replace Disposable Plastic Cups

Submitted by-

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Certificate

This is to certify that the project titled "Sustainable UV-C Hygiene Chamber to Replace Disposable Plastic Cups" is a work carried out by **Pratham Patel (EC149) , Harit Patel (EC139)**, under my supervision as part of the Term Project, as per the prospectus described by the department.

Date:18/11/2025

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Finally, we extend our heartfelt appreciation to one another—**Pratham Patel** and **Harit Patel**—for the collaborative effort, consistent dedication, and teamwork that contributed to the successful completion of this project.

Abstract

The excessive use of disposable plastic cups has become a significant environmental concern, contributing heavily to non-biodegradable waste. Although reusable glass cups offer an eco-friendly alternative, hygiene concerns often discourage their regular use. To address this challenge, we designed a **Sustainable UV-C Hygiene Chamber** capable of efficiently disinfecting glassware without the need for chemicals or complex electronic systems.

The proposed system uses UV-C germicidal light in combination with a fully hardware-based control circuit consisting of dual 555 timers, a relay module, a door safety interlock, and an audible buzzer indicator. The chamber ensures safe operation by activating only when the door is properly closed and by maintaining exposure to UV-C light for an optimal duration to achieve reliable disinfection.

This project aims to promote the hygienic reuse of glass cups in colleges, offices, and public spaces, thereby reducing plastic waste at its source. The system is low-cost, easy to implement, and environmentally sustainable, making it a practical step toward reducing single-use plastics in everyday applications.

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Introduction

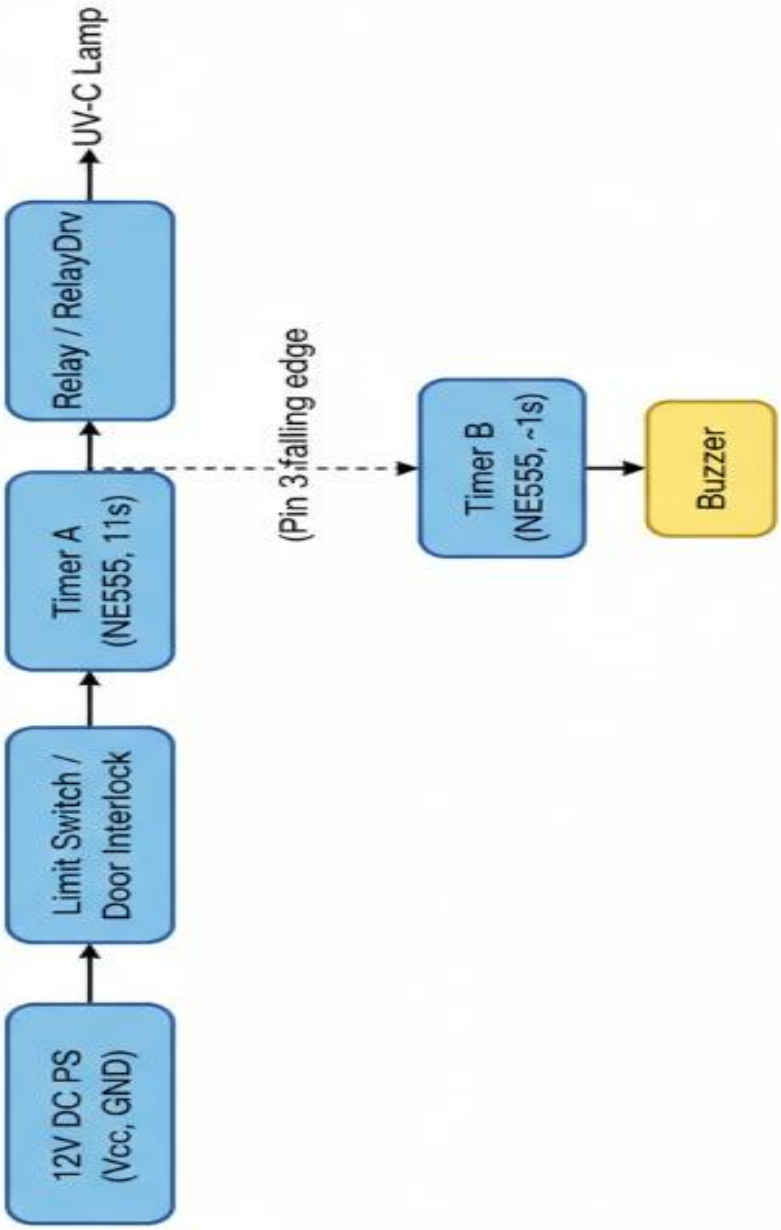
Plastic waste has become one of the most pressing environmental challenges of the modern era. Among the major contributors to this issue are **single-use disposable plastic cups**, widely used in colleges, offices, canteens, and public events due to their convenience and low cost. However, these cups are non-biodegradable, generate significant landfill waste, and pose long-term ecological risks. While reusable glass cups offer a sustainable alternative, concerns regarding hygiene and the risk of bacterial contamination often discourage their widespread adoption.

To bridge this gap between sustainability and hygiene, this project introduces a **Sustainable UV-C Hygiene Chamber**, designed to disinfect reusable glass cups efficiently and safely. UV-C light, known for its germicidal properties, is widely used in medical and laboratory sterilization processes due to its ability to inactivate microorganisms without the use of chemicals. By integrating this technology into a controlled, compact chamber, users can safely disinfect glassware and confidently reuse it.

The project employs a fully hardware-based design using dual NE555 timers, a relay module, and safety interlocks—ensuring reliable operation without the need for microcontrollers or complex programming. The chamber activates only when the door is properly closed, provides timed UV-C exposure, and alerts the user upon completion of the sanitization cycle.

By promoting hygienic reuse of glassware, this system aims to significantly reduce dependency on disposable plastic cups, making it a practical and eco-friendly solution for campuses, workplaces, and community spaces.

Block diagram



Block Diagram Explanation

The block diagram of the *Sustainable UV-C Hygiene Chamber to Replace Disposable Plastic Cups* illustrates the functional flow of the entire system. Each block represents a specific subsystem that contributes to the safe and automated operation of the UV-C disinfection process.

1. Power Supply

The system is powered by a regulated **12 V DC supply**, which provides stable voltage to the NE555 timers, relay module, buzzer, and associated circuitry. Proper decoupling capacitors are used to ensure noise-free operation and to protect sensitive components.

2. Limit Switch (Door Interlock)

A mechanical **limit switch** is installed at the chamber door to act as a safety interlock. The switch ensures that the system only activates when the door is fully closed, preventing accidental exposure to UV-C radiation. When the door closes, the switch provides a trigger signal to Timer A.

3. Timer A (NE555 – UV Exposure Timer)

Timer A is configured in monostable mode to generate a fixed-duration output pulse that controls the UV-C lamp. Upon receiving a trigger from the limit switch, Timer A's output remains HIGH for approximately **11 seconds**, ensuring proper exposure time for effective sterilization. This output controls the relay driver circuit.

4. Relay / Relay Driver

The output of Timer A drives a **relay coil** (directly or via a transistor driver), which acts as a switch for the UV-C lamp. The relay isolates the low-voltage control circuitry from the high-voltage lamp driver, ensuring safety and reliability. The lamp remains ON only during the active period of Timer A.

5. UV-C Lamp

The **UV-C germicidal lamp** provides the required 254 nm wavelength radiation essential for inactivating bacteria, viruses, and microorganisms on the glassware placed inside the chamber. The lamp activates only when the door is closed and Timer A output is HIGH.

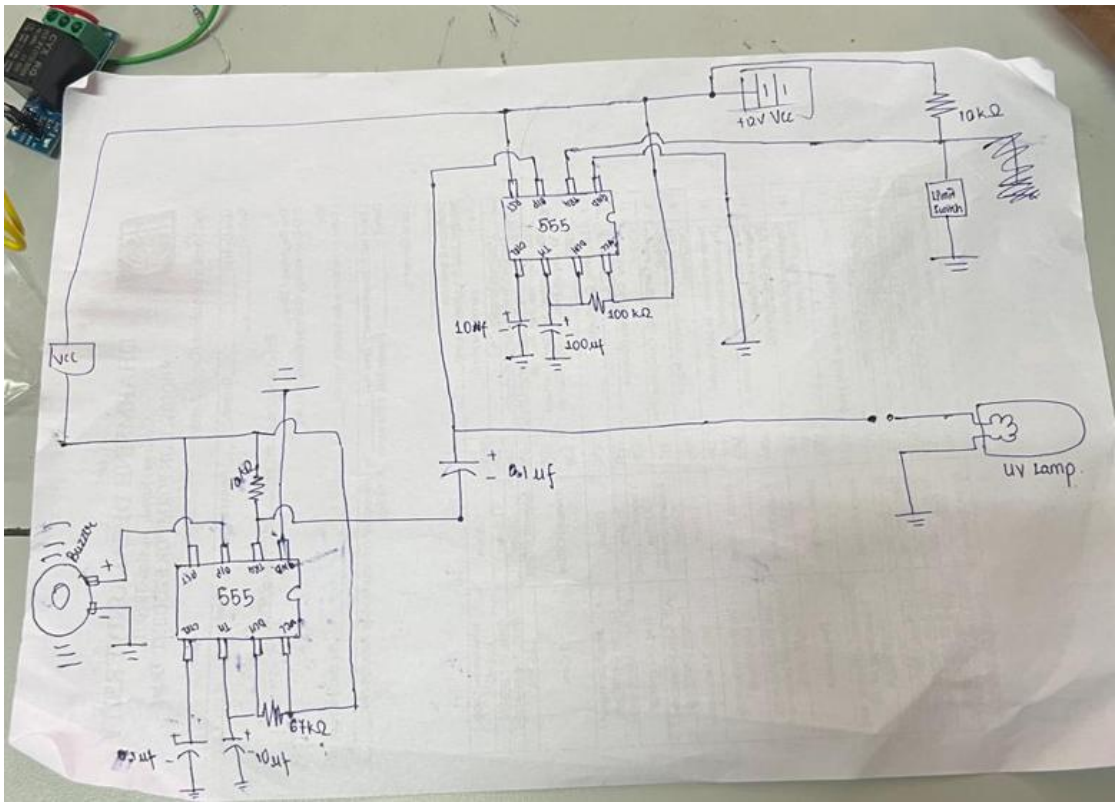
6. Timer B (NE555 – Buzzer Timer)

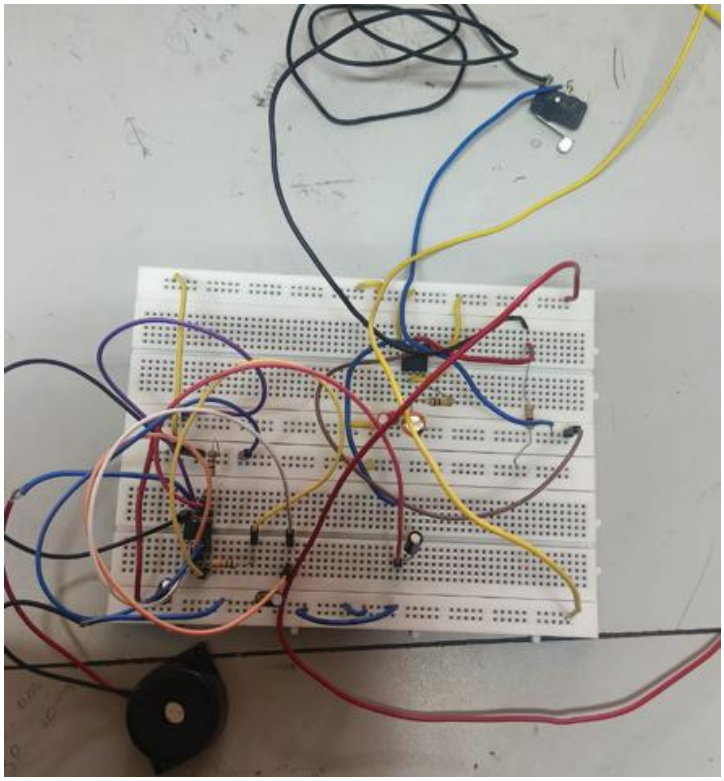
Timer B is also configured in monostable mode, but with a shorter duration (1–2 seconds). It is triggered by the **falling edge** of Timer A's output through a coupling capacitor. This ensures that Timer B activates **only after** the sterilization cycle ends, producing a short alert tone.

7. Buzzer

This stage generates an audible indication to notify the user that the sterilization process is complete.

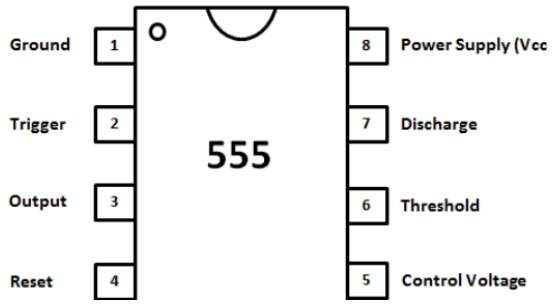
Circuit diagram





Circuit Components

1. **NE555 Timer IC (Texas Instruments / STMicroelectronics Standard Version)**



The NE555 is a precision timing IC capable of generating stable time delays and oscillations. It is used twice in this project (Timer A and Timer B), both configured in monostable mode.

Key Datasheet Specifications

- **Supply Voltage (VCC):** 4.5 V to 16 V
- **Output Current:** up to 200 mA (source or sink)
- **Trigger Voltage (Pin 2):** ≤ 1/3 VCC to activate
- **Threshold Voltage (Pin 6):** ≥ 2/3 VCC to reset timing cycle
- **Reset Pin (Pin 4):** Active LOW
- **Operating Temperature:** 0°C – 70°C

Pin Functions

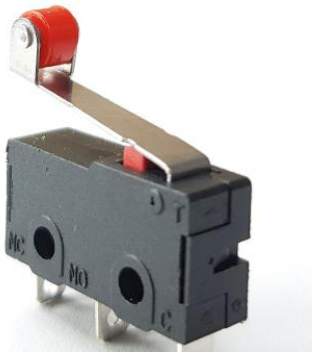
- **Pin 1 (GND):** Reference ground
- **Pin 2 (TRIG):** Activated by LOW pulse
- **Pin 3 (OUT):** Output (HIGH during timing)
- **Pin 4 (RESET):** Disable timing when LOW
- **Pin 5 (CTRL):** Control voltage (optional tuning)
- **Pin 6 (THR):** End timing when reached threshold
- **Pin 7 (DISCH):** Discharges timing capacitor

- **Pin 8 (VCC):** +12 V supply

Why chosen?

- Simple
- Reliable
- Accurate timing
- No programming required

2. Limit Switch (Mechanical Micro-Switch)



Used as door interlock to ensure UV lamp only turns ON when door is closed.

Typical Datasheet Characteristics

- **Contact Type:** SPDT (COM, NO, NC)
- **Operating Force:** ~0.5–1 N
- **Electrical Rating:** 5 A at 250 VAC
- **Bounce Time:** < 10 ms
- **Mechanical Life:** 1 million operations
- **Insulation Resistance:** >100 MΩ

Why chosen?

- High reliability
- Fast switching

- Essential safety feature

3.Capacitors and Resistors (Timing & Filtering Network)

Electrolytic Capacitors (10 µF, 25 V)

- Used for timing (RC constant)
- Tolerance: ±20%
- Leakage: low (but higher than film capacitors)

Ceramic Capacitors (100 nF, 10 nF)

- Used for noise suppression & debouncing
- Tolerance: ±10%
- Very fast response time

Resistors

- **1 MΩ** for Timer A (11s delay)
- **100 kΩ** for Timer B (1–2s delay)
- **1 kΩ** base resistor for BC547
- **10 kΩ** pull-ups for trigger pins

Working Principle

The working principle of the *Sustainable UV-C Hygiene Chamber* is based on controlled ultraviolet (UV-C) exposure, safety interlocking, and timed hardware automation using dual NE555 timer circuits. The system ensures that the UV-C lamp operates only under safe conditions and for a fixed duration required to disinfect reusable glass cups effectively.

The operation begins when the user closes the chamber door. A **limit switch** placed at the door acts as a safety interlock. Only when the door is fully closed does the switch activate, sending a trigger signal to **Timer A**, which is configured in monostable mode. This configuration ensures that upon receiving a LOW trigger, Timer A produces a single, stable output pulse for a predetermined duration. In this project, Timer A maintains its output HIGH for approximately **11 seconds**, representing the UV exposure time required for effective sterilization.

When Timer A output turns HIGH, it energizes the **relay module**, which switches the UV-C lamp ON. During this 11-second interval, the UV-C lamp emits germicidal light at a wavelength around 254 nm, capable of inactivating microorganisms on the glass surface. The chamber remains securely closed during this period, preventing any direct exposure to the user.

Once the timing cycle ends, Timer A's output transitions from HIGH to LOW. This **falling edge** is used to trigger **Timer B**, which is also configured in monostable mode but with a much shorter duration (about 1–2 seconds). Timer B briefly activates a **buzzer** through a transistor driver stage, providing an audible alert to indicate successful completion of the sterilization process.

Throughout the process, the timers, capacitors, pull-up networks, and safety switch ensure noise-free, reliable triggering and safe operation. The design eliminates the need for microcontrollers or software, relying entirely on robust hardware timing principles.

Thus, the system operates automatically, safely, and efficiently:

- Door closes → Timer A triggers → UV-C lamp ON for set duration
- Timer A finishes → Timer B triggers → Buzzer beeps to notify completion

This simple yet effective principle supports hygienic reuse of glassware, helping reduce dependence on disposable plastic cups.

Advantages

1. Environmentally Friendly

Promotes the reuse of glass cups and significantly reduces plastic waste by replacing disposable cups with a safe, hygienic alternative.

2. Fully Automated and Easy to Use

Once the door is closed, the system automatically handles UV sterilization and provides an audible alert when the cycle is complete.

3. Safe Operation with Door Interlock

The limit switch ensures the UV-C lamp operates only when the chamber is closed, preventing accidental UV exposure.

4. Low-Cost Hardware-Only Design

Uses simple components like NE555 timers, relay, and passive parts—making the system inexpensive, reliable, and easy to build or repair.

Limitations

1. Limited Sterilization Area

The chamber can disinfect only small items like glass cups; larger objects cannot be accommodated.

2. UV-C Works Only on Direct Surfaces

UV-C light cannot disinfect surfaces in shadow or covered areas, leading to incomplete sterilization if objects are improperly placed.

3. Fixed Timing (No Digital Adjustment)

Since the system is hardware-based, exposure duration cannot be changed without modifying resistor or capacitor values.

4. UV-C Lamp Degradation Over Time - The sterilization effectiveness gradually reduces as the UV-C lamp ages and requires periodic replacement.

Applications

1. Educational Institutions

Can be installed in college canteens, hostels, and staff rooms to sterilize reusable glass cups, reducing plastic waste across campus.

2. Offices and Workplaces

Useful for disinfecting shared office glassware, especially in pantry areas where hygiene and sustainability are priorities.

3. Cafeterias and Small Food Outlets

Provides a quick and cost-effective method to sanitize cups and small utensils without chemicals or bulky equipment.

4. Homes and Personal Use

Ideal for individuals who want an easy and safe method to sterilize glassware, baby bottles, or small household items.

Future Scope

As the need for sustainable hygiene solutions grows across institutions and public spaces, the *Sustainable UV-C Hygiene Chamber* offers several promising opportunities for expansion and enhancement. Although the current prototype successfully demonstrates safe and automated UV sterilization using a hardware-only approach, the system can be further developed to improve efficiency, safety, user experience, and scalability. The following points highlight possible future advancements:

1. Microcontroller-Based Automation

The current hardware-only design can be upgraded to a microcontroller-based system (Arduino/ESP32) enabling features such as:

- Adjustable sterilization timings
- Real-time LCD display
- Error detection and status monitoring

This improves precision and flexibility.

2. Safety Enhancements

Additional safety features can be implemented to improve user protection:

- Door-open UV automatic shutdown
- UV leakage sensors
- Temperature monitoring inside the chamber

These make the system more suitable for commercial and institutional use.

3. Multi-Object Sterilization

The chamber can be redesigned to accommodate multiple cups or small utensils at once, increasing efficiency and making it practical for cafeterias and restaurants.

4. IoT Integration

By incorporating wireless connectivity, the system can:

- Log cycle counts
- Send completion notifications
- Provide maintenance alerts (e.g., UV lamp nearing end of life)

This enables smart monitoring in large facilities.

With these developments, the UV-C hygiene chamber has the potential to evolve into a scalable, intelligent, and highly efficient sterilization system. Such improvements can broaden its applicability, enhance user convenience, and further support sustainability efforts by promoting hygienic and repeated use of glassware.

Problems faced during the Project

• Incorrect Triggering of Timer B

Initially, Timer B stayed active even when the UV cycle was not running, causing the buzzer to beep at the wrong times. This was due to Timer B receiving a constant LOW level instead of a falling-edge pulse.
Solution: A 100 nF series capacitor and 10 kΩ pull-up were added, converting Timer A's falling edge into a clean, short trigger pulse for Timer B.

• Noise and False Triggering on Trigger Pins

The NE555 trigger pins were extremely sensitive to noise. Mechanical bounce from the limit switch and small fluctuations on the supply caused unexpected self-triggering of Timer A.
Solution: Debounce capacitors (100 nF) and stable pull-up resistors (10 kΩ) were added to both Timer A and Timer B trigger pins, eliminating false pulses completely.

• Safety Concerns with UV-C Lamp Operation

Initial prototype lacked a reliable safety cutoff. The lamp could power ON even when the chamber door was slightly open, posing a risk of UV-C exposure to the user.
Solution: A properly placed door limit switch was installed to allow triggering only when the door is fully closed, ensuring safe operation throughout the sterilization cycle.

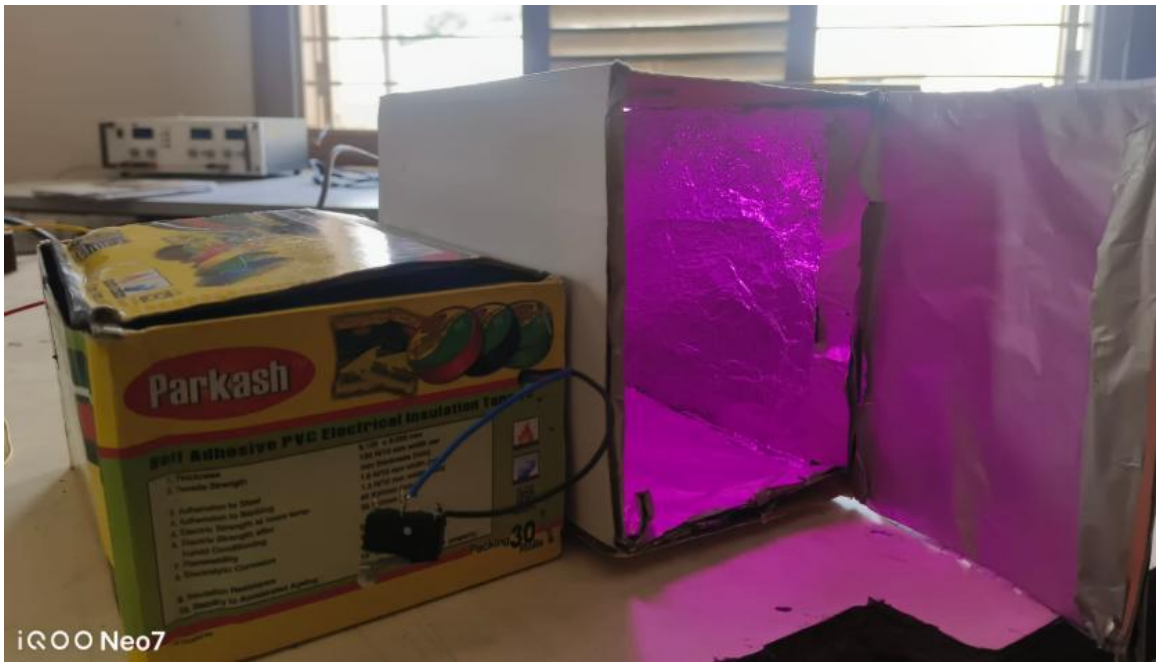
• Timing Inaccuracy Due to Component Tolerances

The UV exposure time varied slightly during early tests because of tolerance variations in electrolytic capacitors and high-value resistors.
Solution: More stable RC values were selected improving timing consistency.

• Loose Breadboard Connections and Wiring Errors

Frequent wiring changes, dual timers, and multiple capacitors caused clutter and intermittent failures on the breadboard.
Solution: The circuit layout was reorganized, wires were color-coded, and ground reference points were grouped, making the system stable and easier to troubleshoot.

Final images of working project



Conclusion

The *Sustainable UV-C Hygiene Chamber* successfully demonstrates a practical and eco-friendly solution to reduce the growing dependence on disposable plastic cups. By utilizing UV-C germicidal light and a fully hardware-based control system built around dual NE555 timers, the project achieves safe, reliable, and chemical-free sterilization of reusable glassware. The inclusion of a door-activated limit switch ensures user safety, while the buzzer alert and automated timing provide a simple, intuitive user experience without requiring any microcontroller programming.

Through this project, we were able to integrate concepts of electronics, safety, sustainability, and real-world problem solving. The iterative troubleshooting—such as correcting relay logic, stabilizing timer triggers, and ensuring proper switching—strengthened our understanding of practical circuit design beyond theoretical knowledge. The final system is cost-effective, easy to construct, and highly adaptable, making it suitable for colleges, offices, cafeterias, and small food outlets aiming to minimize plastic waste.

Overall, this project not only fulfills its technical objectives but also supports a broader environmental goal by encouraging the hygienic reuse of glassware. With further enhancements and scaling, this chamber has the potential to contribute meaningfully to sustainable practices in everyday life.